Serenade® Optimum

Contains: not less than 1.46×10^{10} cfu/g of *Bacillus subtilis* (strain QST 713), in the form of a wettable powder.

Net Contents: **5 kg, 10 kg**

Fungicide

A Bio-Fungicide to aid in the control and suppression of Powdery Mildew, Botrytis and Sour Rot on Grapes, Fire Blight on Pipfruit, Botrytis on Berryfruit, Kiwifruit, Persimmons and Onions, and various diseases on Avocados, Citrus, Lettuce and Turf.

Registered pursuant to the ACVM Act 1997, No P9409
See www.foodsafety.govt.nz for registration conditions.

Approved pursuant to the HSNO Act 1996, No HSR000656
See www.epa.govt.nz for approval conditions.

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WARNING
KEEP OUT OF REACH OF CHILDREN

Causes serious eye irritation.
May cause an allergic skin reaction.
Harmful to terrestrial invertebrates

PRECAUTIONS:

Read label before use
Do not eat, drink or smoke while using this product.
Remove protective clothing and wash hands and face thoroughly before meals and after work.
Avoid contact with the skin and eyes
Take all reasonable steps to ensure that the substance does not cause any significant adverse effects to the environment beyond the application area

PERSONAL PROTECTION:

When handling, wear full protective clothing such as eye protection, impervious gloves, respirator, waterproof hat, coat and trousers (worn outside rubber boots).
STORAGE: Store in original container tightly closed and in a locked dry, cool, well-ventilated area, away from foodstuffs.
SPILLS AND DISPOSAL: When dealing with spills wear personal protective clothing and equipment as described in the PERSONAL PROTECTION section.
Contain the spill by damming, recover spilt product by absorbing with sawdust or an inert absorbent material then transfer the recovered spilt material to a properly labelled drum. Deal with all spillages immediately. If contamination of drains, streams, watercourses, etc. is unavoidable, warn the local water authority.
Disposal: Dispose of this product only by using according to the label or at an approved landfill. Avoid contamination of any water supply with product or empty container.
Container Disposal: Triple rinse container and add residue to mixture. Take to an Agrecovery collection site.

FIRST AID:
If swallowed, do NOT induce vomiting. For advice, contact the National Poisons Centre 0800 POISON (0800 764 766) or a doctor immediately. If skin contact occurs, remove contaminated clothing and flush skin with running water. If splashed in eyes, wash out immediately with water, if irritation persists get medical advice. If inhaled, move the victim to fresh air immediately.

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GENERAL INFORMATION;
Serenade Optimum has been developed from naturally-occurring strains of *Bacillus subtilis*. It is a broad spectrum preventative fungicide recommended for the control of Powdery Mildew, Grey Mould (*Botrytis* sp.) and Sour Rot in grapes during periods of low disease infection pressure. It acts by preventing spore germination, germ tube elongation and germ tube penetration. Serenade Optimum can be integrated into an overall disease and pest management strategy wherever fungicide use is necessary in an IPM programme and is also an important tool in resistance management strategies. Serenade Optimum should be applied prior to or in the early stages of disease development.

RESISTANCE MANAGEMENT:

Group	44	Fungicide

Serenade Optimum contains *Bacillus subtilis* strain QST 713 from the Microbial (*Bacillus* sp.) from the Group 44 fungicide group.
This category was established by FRAC to accommodate the unique properties of *Bacillus subtilis* strain QST 713. With no known resistance issues and a low risk assumption based on the mode of action, Serenade Optimum will assist in managing fungicide resistance provided it is used strictly in accordance with the label instructions and resistance management strategies that exist for any fungus listed on the label.

It is an offence to use this product on animals.

DIRECTIONS FOR USE:
Thorough coverage is essential for optimum disease control. The rates recommended are for high volume spraying to the point of run off. For concentrate spraying, adjust dilution rate accordingly.

Crop	Disease	Remarks
Note: For the control of any disease, use alternative fungicides when disease pressure is high and during the rest of the season prior to harvest.		
Apples, Pears	Fire blight	Make the first application up to 5% blossom and apply 125-250 g/100 litres (or 1.25 - 2.5 kg/ha) following a streptomycin application to minimise disease resistance to antibiotics. Under severe disease infection periods, repeat applications may be made at 3-7 day intervals, using the longer interval after petal fall. The addition of a suitable adjuvant (e.g. Du-Wett®) is recommended.
Avocados	Anthracnose	Apply 100 g/100L in a tank mixture with the lowest rate of a copper-based fungicide, or 150 g /100 litres alone over flowering.
The use rates for all subsequent crops are 1.25 2.0 kg/ha or 125-200 g/100 litres in sufficient water to ensure complete coverage of plants and flower parts. The lower rates are generally for preventative applications and higher rates for conditions favouring disease infection.		
Citrus	Botrytis	Apply at 50% flowering and repeat at 7-10 day intervals if disease conditions persist. Ensure high volume application with surfactant for coverage.
	Anthracnose	Apply at monthly intervals from early flowering to harvest.
	Greasy spot	For suppression, begin applications at first leafy flush, and repeat at later flushes. Include spray oil in tank mixture. Can also be tank mixed with other fungicides including copper-based ones.
	Melanose	For suppression, begin applications at petal fall and repeat at 14-21 day intervals.
Grapes	Powdery mildew, Botrytis, Sour rot	Serenade Optimum can be used all season, particularly in organic crops. Apply as a preventative application at key growth stages. Under high disease pressure in conventional blocks, use the best available alternatives at critical timings 80% cap fall and bunch closure. Serenade Optimum can be used later in the season as it has nil residues and no withholding period restrictions. When using as a pre harvest preventative spray, use 2 to 3 applications at 7 to 10 day intervals, starting 4 to 6 weeks from harvest. Use the higher rate

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		and shorter interval when conditions favour rapid disease development. Apply in sufficient water to ensure thorough coverage of all parts of the plants early to mid-season and, in late season applications, target the bunch line area.
Kiwifruit	Botrytis	Apply when disease conditions are conducive to infection at late flowering to early fruit set and immediately pre-harvest. Under heavy disease pressure, use in a spray program with other fungicides. May give suppression of Sclerotinia.
Persimmons	Botrytis	Apply over flowering and repeat at 7-10 day intervals if disease conditions persist.
Strawberries Berryfruit	Botrytis	Apply from early flowering and during harvest at 10-14 day intervals if disease conditions persist. Include a wetting agent and repeat in a program of alternation with other fungicides.
Lettuce	Sclerotinia minor	Does NOT control Sclerotiorum. Apply as a soil drench immediately before or after transplanting or seedling crop emergence. Repeat as necessary before crop coverage of the soil and alternate with another fungicide for greater disease control spectrum.
Onions	Botrytis, Downy mildew	As part of a program, apply pre-harvest for the moderate control and suppression of diseases
Ornamentals	Botrytis, Powdery mildew	Apply at the first sign of disease and repeat at 7-10 day intervals.
Turf	Anthracnose, Brown patch, Dollar spot, Powdery mildew, Rust	Apply when conditions are conducive to disease development. Use the higher rate for improved control and the lower rate when in tank mixture with a suitable fungicide or in a rotation program with alternative chemistry. Repeat as necessary at 7-10 day intervals. The addition of Du-Wett® adjuvant is strongly recommended.

Mixing:
Partially fill the spray tank with clean water and start agitation. Add the specified amount for the area to be treated of Serenade Optimum to the spray tank. Complete filling the tank to the required volume for the area to be treated. Maintain agitation until spraying is complete. Do NOT allow the spray mixture to stand overnight or for any prolonged periods. Spray pH must be not less than pH 4.5 and not greater than pH 8.5.

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Compatibility:
Do NOT combine Serenade Optimum in the spray tank with pesticides, surfactants, copper-based products or fertilisers unless prior experience has shown the combination to be compatible. Any tank combinations must also be tested for effectiveness on disease and selectivity to plants before use on commercial areas.
If the use of any tank mixture is desirable, a small volume of the mixture should first be tested in a small measured volume, for physical compatibility at the dilution rates required. If this test is positive, a field test of the mixture should then be carried out on a few plants to observe effectiveness and plant safety.

Application:
Serenade Optimum is intended for use as a dilute spray in water for application through conventional ground spray equipment. Thorough coverage is essential for optimum disease control. To achieve good coverage, use of the correct spray pressure, water volume nozzle-size and spacing with the correct ground speed are critical. Sprayers should be calibrated before use. To do this, consult spray nozzle catalogues and other industry references for specific information on sprayer calibration or contact a registered 'sprayer calibration' consultant.

It is an offence for users of this product to cause residues exceeding the relevant MRL in the Food Notice: Maximum Residue Levels for Agricultural Compounds

Leaf-plucking in vineyards: Sheep must not be sent for slaughter or milked for 2 months after being removed from the vineyard (and placed on clean feed)

CONDITIONS OF SUPPLY:
READ THESE CONDITIONS OF SUPPLY BEFORE BUYING AND USING. IF THESE CONDITIONS ARE UNACCEPTABLE, RETURN AT ONCE UNOPENED.

License and Prohibition of Re-Sale: Bayer CropScience LP hereby grants buyer or user the right under the patents listed on the label to use this product solely in accordance with the label's Directions for Use for applications to plants, including plant parts such as seed, or to soil, where the treated plants or the plants grown in treated soil are intended for sale, in whole or in part, or are intended for public or personal use. The buyer or user does not have the right to de-formulate this product or to isolate and/or culture its active ingredient for any purpose. Unless specifically granted in writing, the buyer or user does not have the right to re-sell this product in any form; e.g., this product may not be re-sold in combination with other products or other active ingredients or in a diluted form, unless combinations are prepared and delivered to the end-user for immediate application to plants, plant parts or soil solely in accordance with the label Directions for Use.

To the extent permitted by law Bayer does not accept any liability or responsibility whatsoever for any loss, damage or injury whether direct, indirect or consequential howsoever arising in connection with these goods, including without limitation their storage, application, handling or use. Bayer does not make any warranties (express or implied) with respect to fitness for purpose, merchantable quality or any other matter. To the extent only that Bayer is prevented by law from contracting out of the Consumer Guarantees Act, then these conditions apply subject to that Act. The Act shall not apply where these goods are acquired for business purposes.

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Date of Manufacture:

Batch No:

Serenade® Optimum is a registered trademark of Bayer CropScience LP
Du-Wett® is a registered trademark of Elliott Technologies Ltd.

Bayer

Registered to:
Bayer New Zealand Limited
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